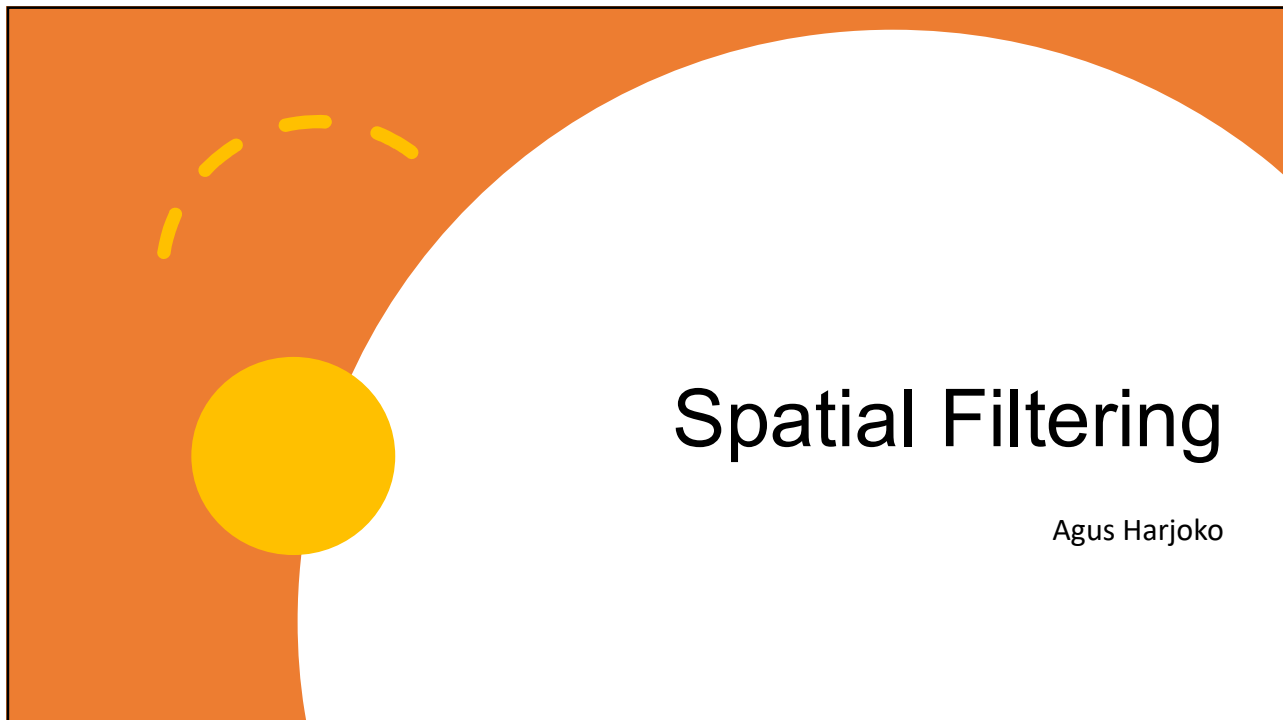
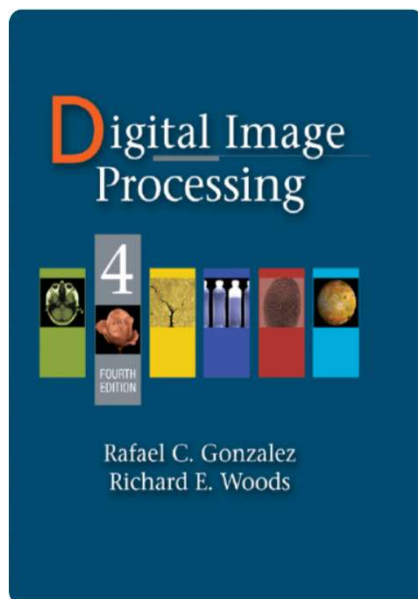




1



Source:

Chapter 3: Intensity Transformations and Spatial Filtering

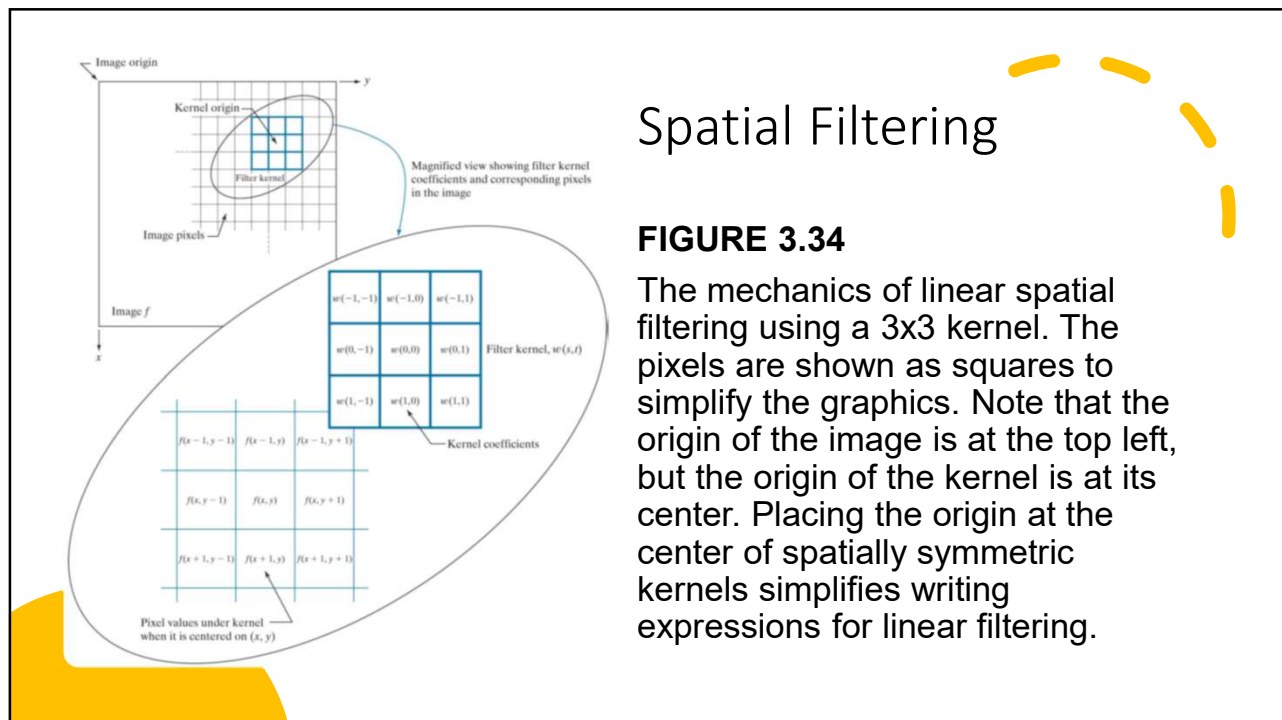
- Title: Digital Image Processing
- Authors:
 - Rafael C. Gonzalez, University of Tennessee,
 - Richard E. Woods, Interapptics.
- Publisher: New York, NY : Pearson, [2018]

2

Objectives

- Understand the mechanics of spatial filtering, and how spatial filters are formed.
- Understand the principles of spatial convolution and correlation.
- Be familiar with the principal types of spatial filters, and how they are applied.
- Be aware of the relationships between spatial filters, and the fundamental role of lowpass filters.
- Understand the principles of fuzzy logic and how these principles apply to digital image processing.

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Correlation and Convolution

- Correlation

$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x+s, y+t)$$

- Convolution

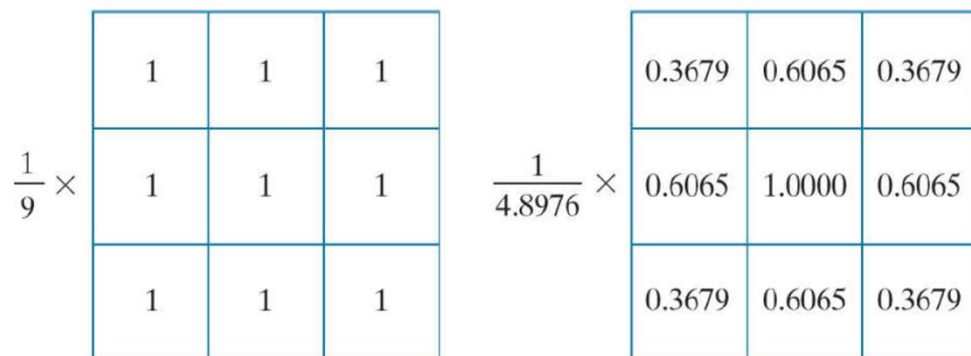
$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x-s, y-t)$$

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Spatial Kernel

FIGURE 3.37

Examples of smoothing kernels: (a) is a *box* kernel; (b) is a *Gaussian* kernel.

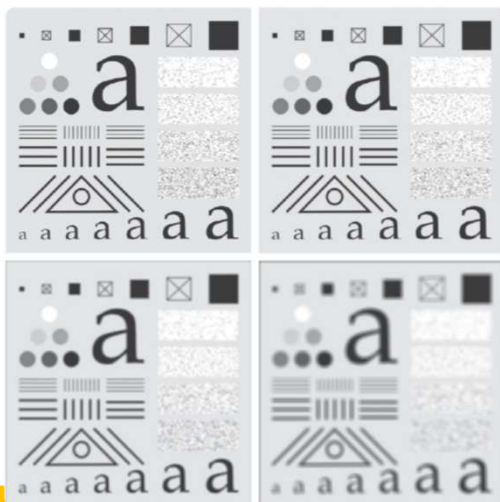


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3.5 Smoothing (Lowpass) Spatial Filters

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Lowpass Box Filter Kernel

FIGURE 3.39

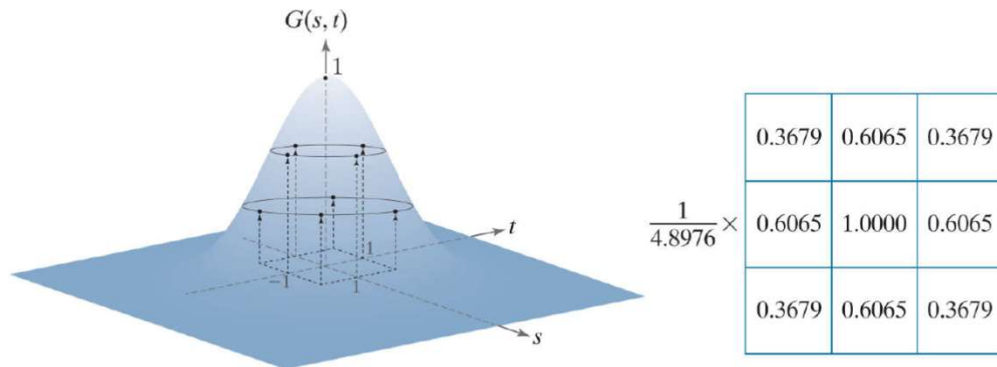
(a) Test pattern of size 1024 x 1024 pixels. (b)-(d) Results of lowpass filtering with box kernels of sizes 3 x 3, 11 x 11 and 21 x 21 respectively.

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Lowpass Gaussian Filter Kernel

FIGURE 3.41

(a) Sampling a Gaussian function to obtain a discrete Gaussian kernel. The values shown are for $K = 1$ and $\sigma = 1$. (b) Resulting 3×3 kernel [this is the same as **Fig. 3.37(b)**]

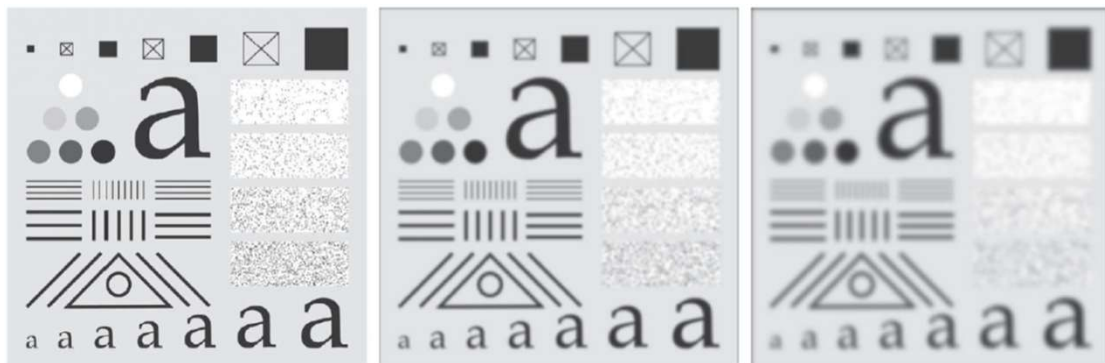


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Smoothing with Gaussian Filter

FIGURE 3.42

(a) A test pattern of size 1024×1024 . (b) Result of lowpass filtering the pattern with a Gaussian kernel of size 21×21 with standard deviations $\sigma = 3.5$. (c) Result of using a kernel of size 43×43 with $\sigma = 7$. This result is comparable to **Fig. 3.39(d)**. We used $K = 1$ in all cases.

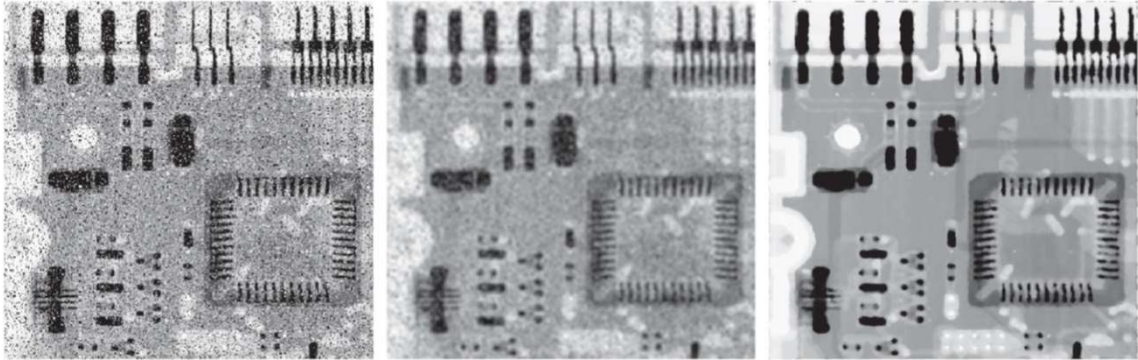


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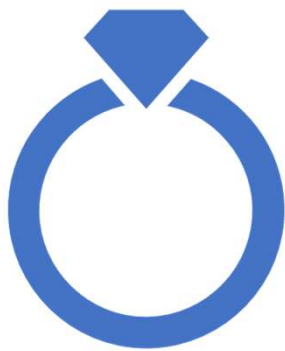
Median Filtering

FIGURE 3.49

(a) X-ray image of a circuit board, corrupted by salt-and-pepper noise. (b) Noise reduction using a 19×19 Gaussian lowpass filter kernel with $\sigma = 3$. (c) Noise reduction using a 7×7 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)



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3.6 Sharpening (Highpass) Spatial Filters

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Laplacian Kernel

FIGURE 3.52

(a) Blurred image of the North Pole of the moon. (b) Laplacian image obtained using the left most kernel. (c) Image sharpened using **Laplacian equation (3.63)** with $c = 1$. (d) Image sharpened using the same procedure, but with the second left most kernel (Original image courtesy of NASA.)

0	1	0	1	1	1	0	-1	0	-1	-1	-1
1	-4	1	1	-8	1	-1	4	-1	-1	8	-1
0	1	0	1	1	1	0	-1	0	-1	-1	-1



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Using First-Order Derivatives for Image Sharpening—the Gradient

FIGURE 3.56

(a) A 3 x 3 region of an image, where the z s are intensity values. (b)–(c) Roberts cross-gradient operators. (d)–(e) Sobel operators. All the kernel coefficients sum to zero, as expected of a derivative operator

	z_1	z_2	z_3	
	z_4	z_5	z_6	
	z_7	z_8	z_9	
	-1	0	0	-1
	0	1	1	0
-1	-2	-1	-1	0
0	0	0	-2	0
1	2	1	-1	0

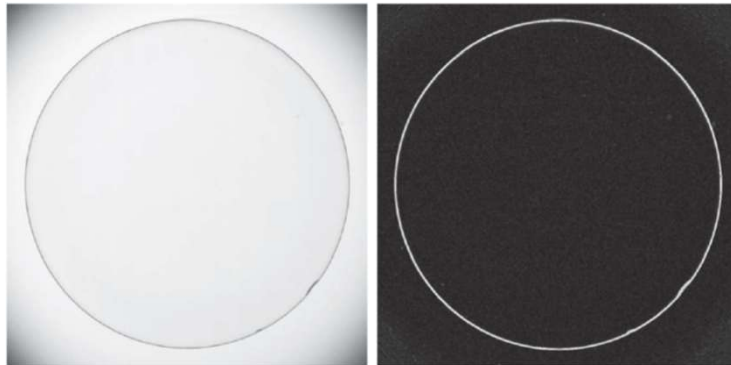
$$\nabla f \equiv \text{grad}(f) = \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

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Gradient for Edge Enhancement

FIGURE 3.57

(a) Image of a contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Perceptics Corporation.)

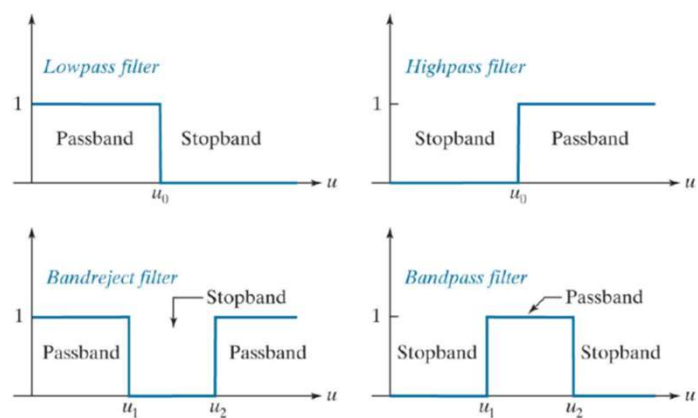


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3.7 Highpass, Bandreject, and Bandpass Filters from Lowpass Filters

FIGURE 3.58

Transfer functions of ideal 1-D filters in the frequency domain (u denotes frequency). (a) Lowpass filter. (b) Highpass filter. (c) Bandreject filter. (d) Bandpass filter. (As before, we show only positive frequencies for simplicity.)



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TABLE 3.7 Summary of the four principal spatial filter types expressed in terms of lowpass filters. The centers of the unit impulse and the filter kernels coincide.

Filter type	Spatial kernel in terms of lowpass kernel, lp
Lowpass	$lp(x, y)$
Highpass	$hp(x, y) = \delta(x, y) - lp(x, y)$
Bandreject	$br(x, y) = lp_1(x, y) + hp_2(x, y)$ $= lp_1(x, y) + [\delta(x, y) - lp_2(x, y)]$
Bandpass	$bp(x, y) = \delta(x, y) - br(x, y)$ $= \delta(x, y) - [lp_1(x, y) + [\delta(x, y) - lp_2(x, y)]]$

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